

Statistical Arbitrage

Ishan Shah



Part I

Stationarity

Math behind ADF Test

ADF Test

Mean-reversal Strategy

Part II

Cointegration

Cointegration Test

Pairs Trading Strategy

Cointegration vs. Correlation

Pairs Selection

Risk Management

- **Directional trading**

Dependent on single instrument.

Example: Corn Futures, Gold Futures, etc.

- **Pairs trading, triplets and other cointegrated trading**

Relative value of 2, 3 or more instruments.

Example: Google vs. Facebook

Statistical Arbitrage

Stationarity



Daniel Kahneman cited a famous example:

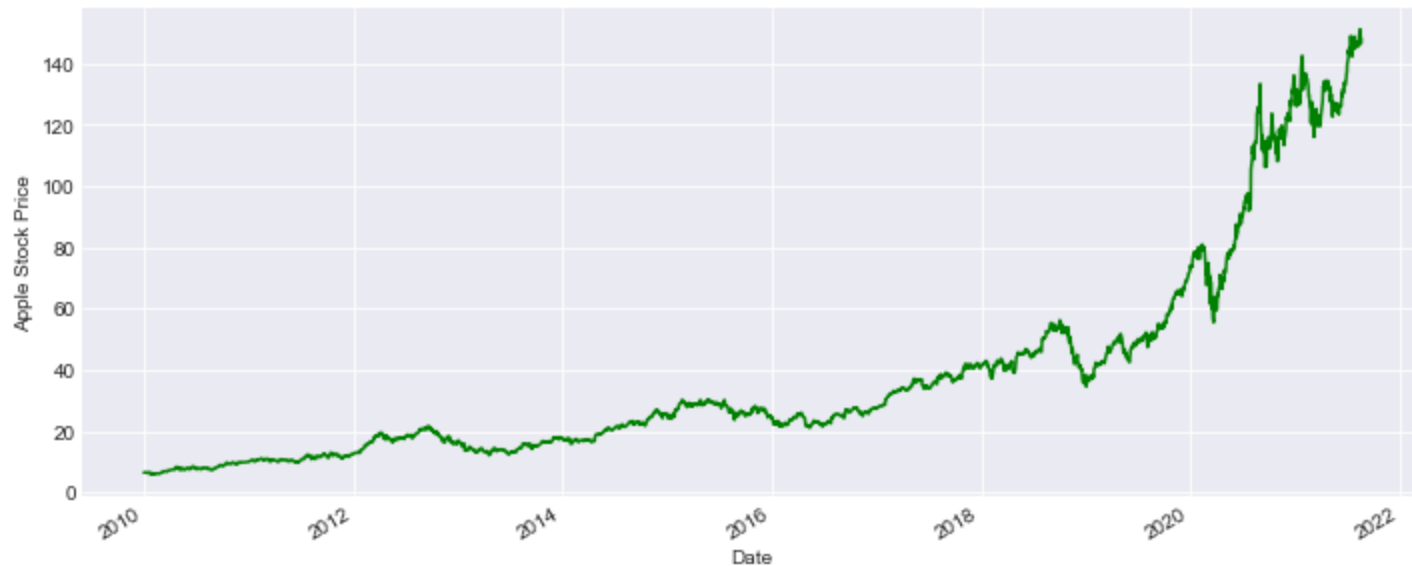
The “Sports Illustrated jinx,” which is the claim that *“an athlete whose picture appears on the cover of the magazine is highly probable to do badly in the upcoming season”*.



The magazine cover for the January 21, 2002 issue of *Sports Illustrated*

Is this stationary?

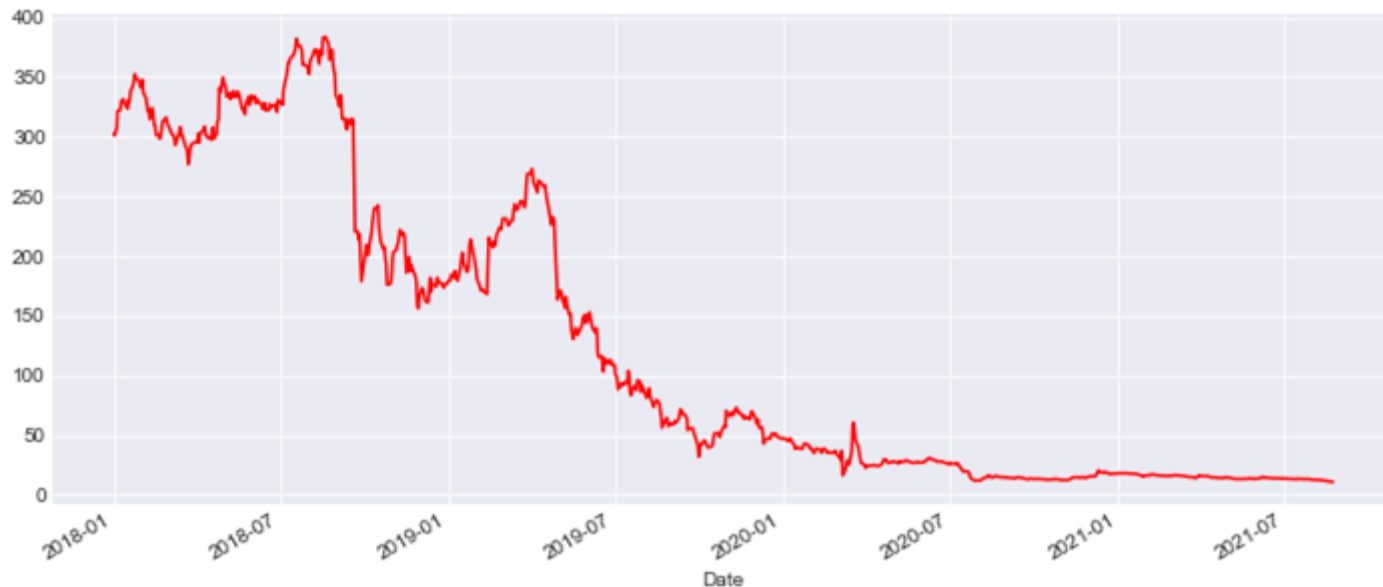
Apple Inc.



Source: Yahoo Finance

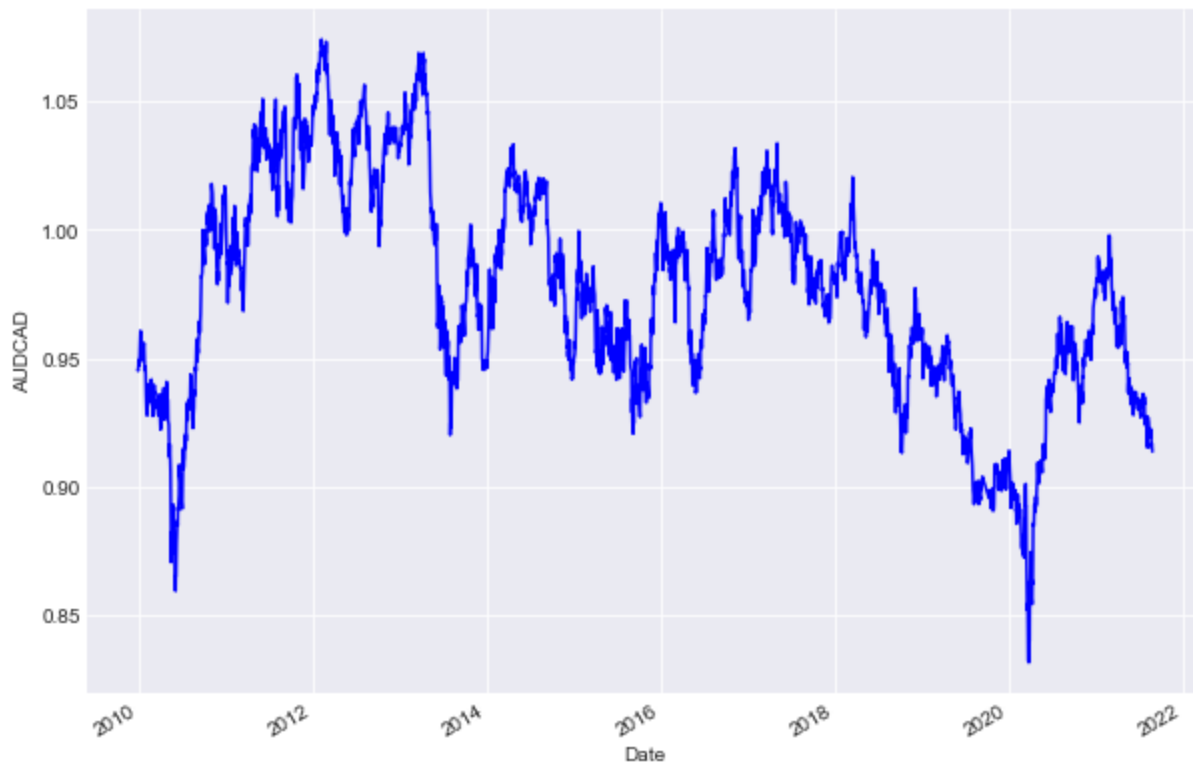
Is this stationary?

Yes Bank



Source: Yahoo Finance

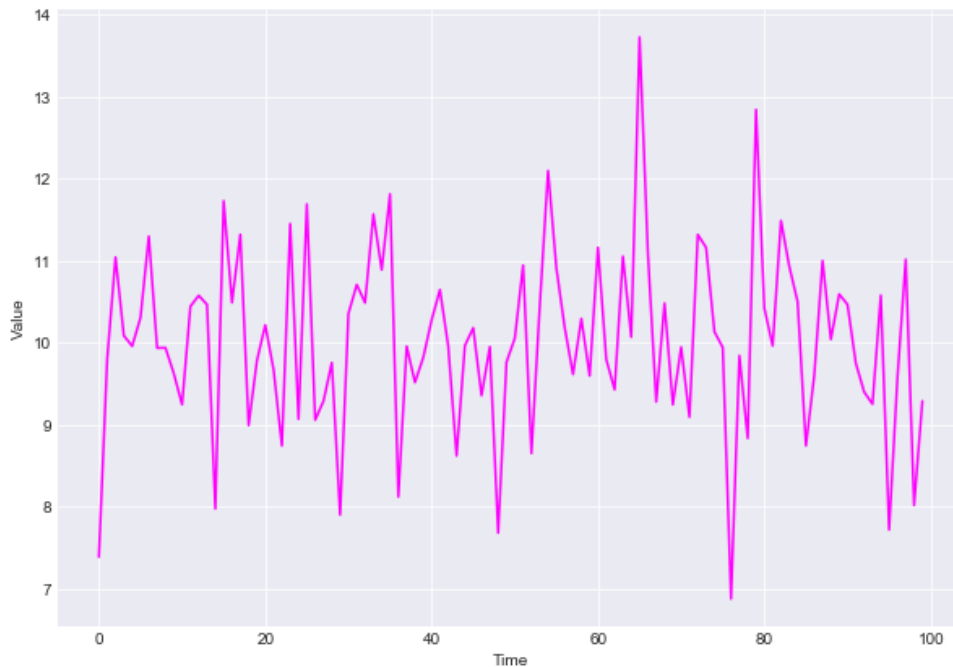
Is this stationary?



Source: Yahoo Finance

What is Stationarity?

A price series is stationary if it doesn't deviate much but stays around the mean.

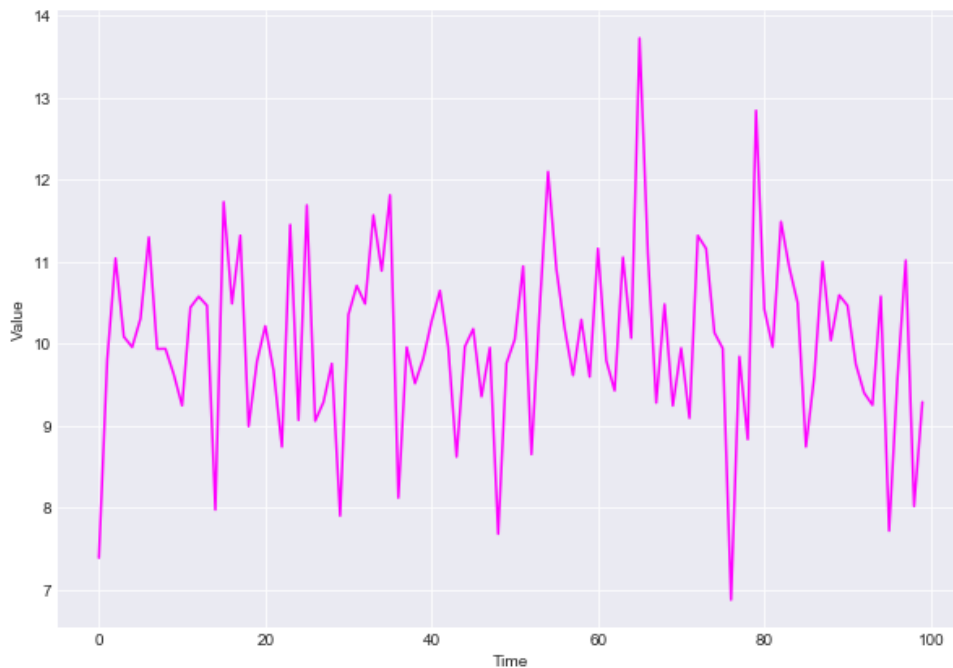


- Random walk is not stationary
- It only applies to long term properties of a price series
- Life is not easy!
- It is difficult to find naturally occurring price series which are stationary

Random walk is not stationary:

[https://www.mit.edu/~kardar/teaching/projects/chemotaxis\(AndreaSchmidt\)/random.htm](https://www.mit.edu/~kardar/teaching/projects/chemotaxis(AndreaSchmidt)/random.htm)

Which Strategy will Work on Stationary Time Series?



Statistical Arbitrage

I DIDN'T HAVE ANY
ACCURATE NUMBERS
SO I JUST MADE UP
THIS ONE.

scottadams@aol.com

STUDIES HAVE SHOWN
THAT ACCURATE
NUMBERS AREN'T ANY
MORE USEFUL THAN THE
ONES YOU MAKE UP.

5809 © 2008 Scott Adams, Inc./Dist. by UFS, Inc.

HOW
MANY
STUDIES
SHOWED
THAT?

EIGHTY-
SEVEN.

ADF Test

\$4,629,873

www.dilbert.com

© Scott Adams, Inc./Dist. by UFS, Inc.

- That's an intuitive definition, but how to check for this statistically?
- There is a convenient statistical test:
 - Augmented Dickey-Fuller (ADF) Test

Can't we just run a backtest on the trading strategy directly and be done with it? Why do we need a statistical test?

- We can describe the current price changes from the past price data using a linear model

$$\Delta p(t) = \lambda p(t-1) + \mu + \beta_t + \alpha_1 \Delta p(t-1) + \dots + \alpha_k \Delta p(t-k) + \epsilon_t$$

p = price of the instrument

$$\Delta p(t) = p(t) - p(t-1),$$

$$\Delta p(t-1) = p(t-1) - p(t-2), \text{ and so on.}$$

λ = regression coefficient

- $H_0: \lambda = 0$ **Not Stationary**

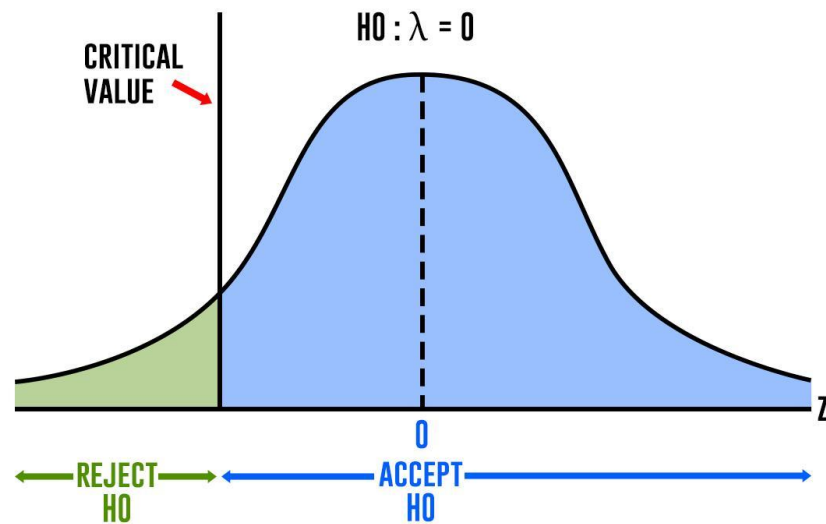
- $H_a: \lambda < 0$ **Stationary**

- Intuition:

If the hypothesis $\lambda = 0$ is rejected -

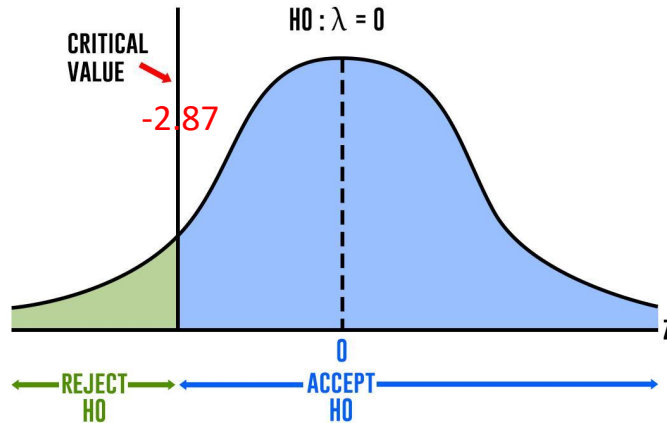
Next move ($\Delta p(t)$) --> Current level ($p(t-1)$)

Hence, it is stationary.



Probability	Critical Value
90%	-2.59
95%	-2.87
99%	-3.44

1. Is t-stat value of -3.05 stationary with 95% confidence level (critical value: -2.87)?
 - Yes/No
2. Is t-stat value of -2.40 stationary with 95% confidence level (critical value: -2.87)?
 - Yes/No



- Read in AUDCAD.csv file

```
data = pd.read_csv('file_name.csv', index_col=0)
```

- Plot the time series

```
data.Close.plot(figsize=(8,4))
```

- Check for Stationarity using adfuller method available in statsmodels.tsa.stattools package

```
help(adfuller)
```

- Set *maxlag* = 1 (Assuming short-range serial correlation)

Perform ADF Test

```
adf = adfuller(data.Close, maxlag = 1)
```

(Advanced topic: To find optimal maxlag, use AIC/BIC.)

- Test statistic: *adf*[0]

- Conclusion:

Since t-stat < -2.59, AUDCAD is stationary with more than 90% certainty

Mean Reversion Strategy



- Read data from CSV file
- Moving average and moving standard deviation
- Upper band and lower band
- Long positions
- Short positions
- Strategy PnL

- Read data from CSV file
- Moving average and moving standard deviation

```
df['moving_average'] = df.Close.rolling(5).mean()
```

```
df['moving_std_dev'] = df.Close.rolling(5).std()
```

- Upper band and lower band

```
df['upper_band'] = df.moving_average + 0.5*df.moving_std_dev
```

```
df['lower_band'] = df.moving_average - 0.5*df.moving_std_dev
```

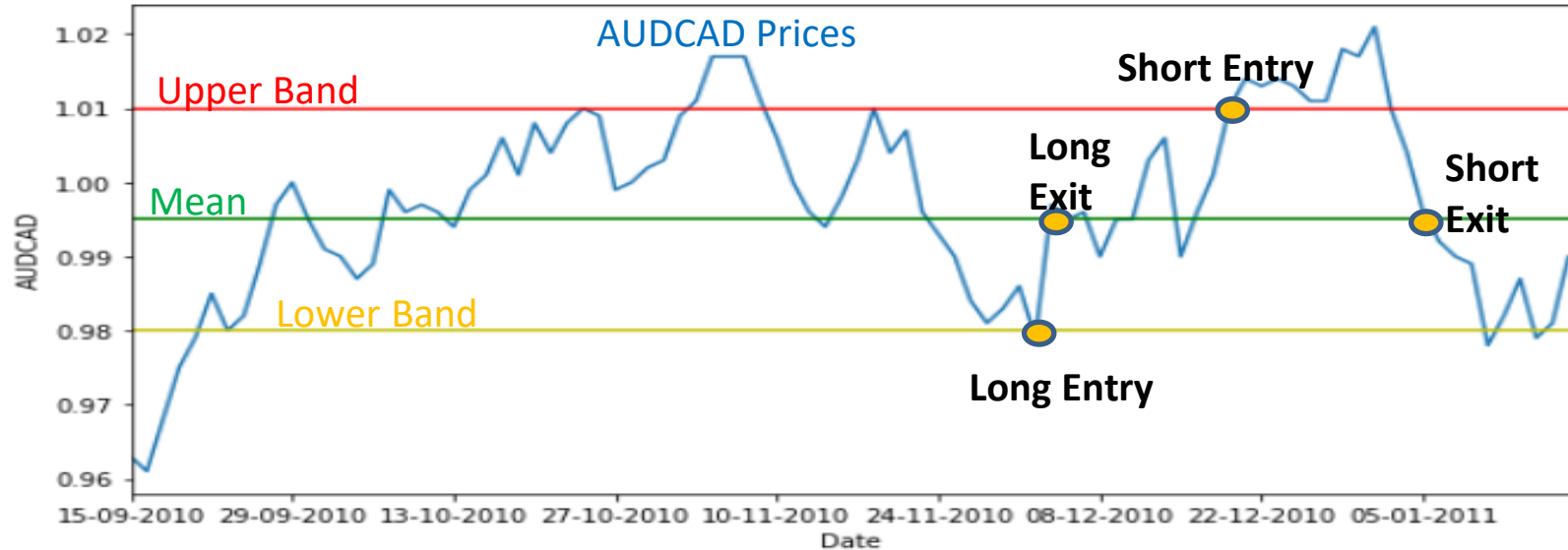
How to select lookback period (5) and width of Bollinger bands (0.5)?

1. Optimize in training set
2. Set lookback to some multiple of half-life

<https://github.com/QuantInsti/EPAT/blob/master/Statistical%20Arbitrage/optimization.py>

https://github.com/QuantInsti/EPAT/blob/master/Statistical%20Arbitrage/half_life.py

Bollinger Bands: AUDCAD



Note: The straight line for mean, upper band and lower band is for illustration purpose only. In the strategy, rolling mean, rolling upper band and rolling lower band is used.

Data source: FXCM

Day	df.Close	df.moving_average	df.lower_band	df.long_entry	df.long_exit
1	98.5	99.5	98	FALSE	FALSE
2	97.5	99.5	98	TRUE	FALSE
3	98.2	99.5	98	FALSE	FALSE
4	99.2	99.5	98	FALSE	FALSE
5	99.6	99.5	98	FALSE	TRUE
6	100	99.5	98	FALSE	TRUE
7	99	99.5	98	FALSE	FALSE

- Long Entry

$df['long_entry'] = df.Close < df.lower_band$

- Long Exit

$df['long_exit'] = df.Close \geq df.moving_average$

df.long_entry	df.long_exit	positions_long
FALSE	FALSE	NaN
TRUE	FALSE	1
FALSE	FALSE	NaN
FALSE	FALSE	NaN
FALSE	TRUE	0
FALSE	TRUE	0
FALSE	FALSE	NaN

- Long Positions

df['positions_long'] = np.nan

df.loc[df.long_entry,'positions_long'] = 1

df.loc[df.long_exit,'positions_long'] = 0

df.long_entry	df.long_exit	positions_long	Forward Filled (positions_long)
FALSE	FALSE	NaN	NaN
TRUE	FALSE	1	1
FALSE	FALSE	NaN	1
FALSE	FALSE	NaN	1
FALSE	TRUE	0	0
FALSE	TRUE	0	0
FALSE	FALSE	NaN	0

- Forward Fill NaN Values

Carry forward an existing position, whenever the next bar's position has not been predetermined to be 0 or 1

```
df.positions_long = df.positions_long.fillna(method='ffill')
```

- **Short Positions**

df['short_entry'] = df.Close > df.upper_band

df['short_exit'] = df.Close <= df.moving_average

df['positions_short'] = np.nan

df.loc[df.short_entry, 'positions_short'] = -1

df.loc[df.short_exit, 'positions_short'] = 0

- **Carry forward an existing position, whenever the next bar's position has not been predetermined to be 0 or -1**

df.positions_short = df.positions_short.fillna(method='ffill')

- **Final Positions**

`df['positions'] = df.positions_long + df.positions_short`

- **Pnl**

`df['prices_difference'] = df.Close - df.Close.shift(1)`

`df['pnl'] = df.positions.shift(1) * df.prices_difference`

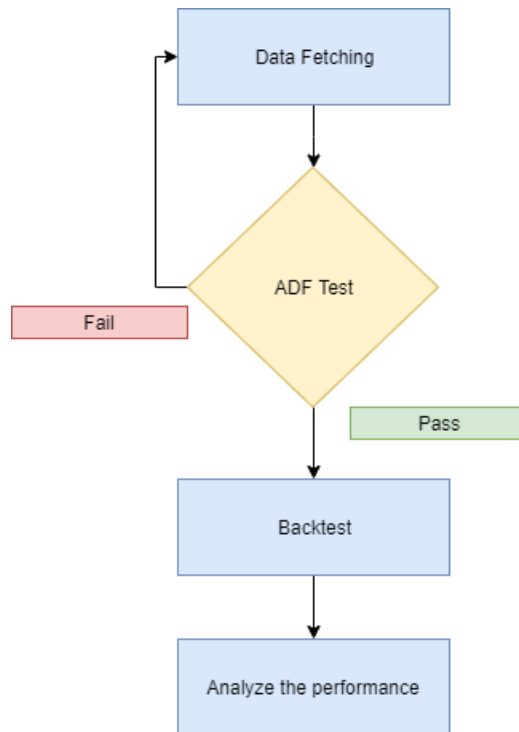
`df['cumpnl'] = df.pnl.cumsum()`

- **Returns**

`df['percentage_change'] = df.Close.pct_change()`

`df['strategy_returns'] = df.positions.shift(1) * df.percentage_change`

`df['cumulative_returns'] = (df.strategy_returns+1).cumprod()`



Pairs Trading

- First developed and used in the mid 1980s by Nunzio Tartaglia's quantitative group at Morgan Stanley.
- Pair Trading is a “*contrarian strategy*” designed to harness mean-reverting behavior of cointegrated instruments.
 - Question: Why Contrarian?
- David Shaw, founder of D.E. Shaw & Co., left Morgan Stanley and started his own “Quant” trading firm in the late 1980's dealing mainly in pair trading.

- A portfolio of two or more instruments such that the portfolio is stationary. Then, the instruments in the portfolio are said to be cointegrated
- Mean Reversion Strategy
 - *Buy the portfolio when its value is low*
 - *Sell the portfolio when its value is high*

- EWA: iShares MSCI Australia ETF

The iShares MSCI Australia ETF seeks to track the investment results of an index composed of Australian equities.

- EWC: iShares MSCI Canada ETF

The iShares MSCI Canada ETF seeks to track the investment results of an index composed of Canadian equities

Source: <https://www.ishares.com/us/products/239607/ishares-msci-australia-etf>

Example of Prices of EWA and EWC



Data source: Yahoo finance

Example of Cointegrated Price Series



$\text{Spread} = \text{EWC} - 1.45 * \text{EWA}$
Strategy – buy low and sell high!

Data source: Yahoo finance

- Any arbitrary combination of the instruments are not cointegrated
- Only some combinations of the instruments are cointegrated
- Linear regression to find the hedge ratio

```
import statsmodels.api as sm
```

```
model = sm.OLS(df.EWC.iloc[:90], df.EWA.iloc[:90])
```

```
model = model.fit()
```

```
print(model.params[0])
```

How to select the lookback period (90 days)?

- Consider two ETFs as X (EWA) and Y (EWC)
 - Hedge Ratio: Linear Regression *X and Y*
 - Spread: $Y - \text{Hedge Ratio} * X$
 - Stationarity Test (ADF test) on the spread
- Test statistic: *adf[0]*
- Conclusion:

Since t-stat < -2.56, EWA and EWC is cointegrated with more than 90% certainty

- **Compute**
 - **Spread: $Y - \text{Hedge Ratio} * X$**
 - **Mean of the Spread**
 - **Standard Deviation of the Spread**
 - **Upper Band and Lower Band**

- Entry conditions

- Long Entry:

- If spread < lower band then

- buy 1 share of Y

- sell hedge ratio share of X

- Short Entry:

- If spread > upper band then

- sell 1 share of Y

- buy hedge ratio share of X

- Exit conditions

- PnL

Cointegration Vs. Correlation

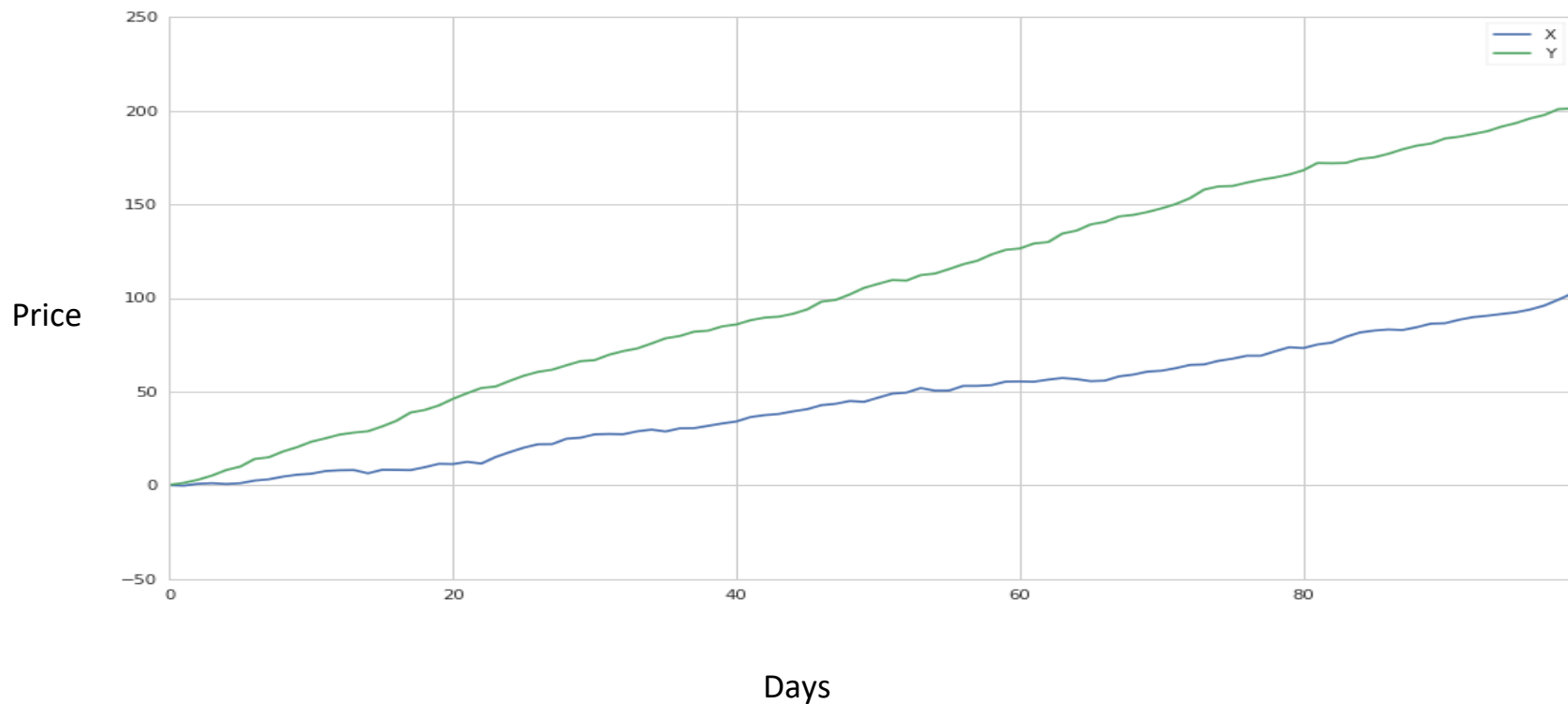
Cointegration

- It is whether the spread between two instruments is stationary
- Long-term

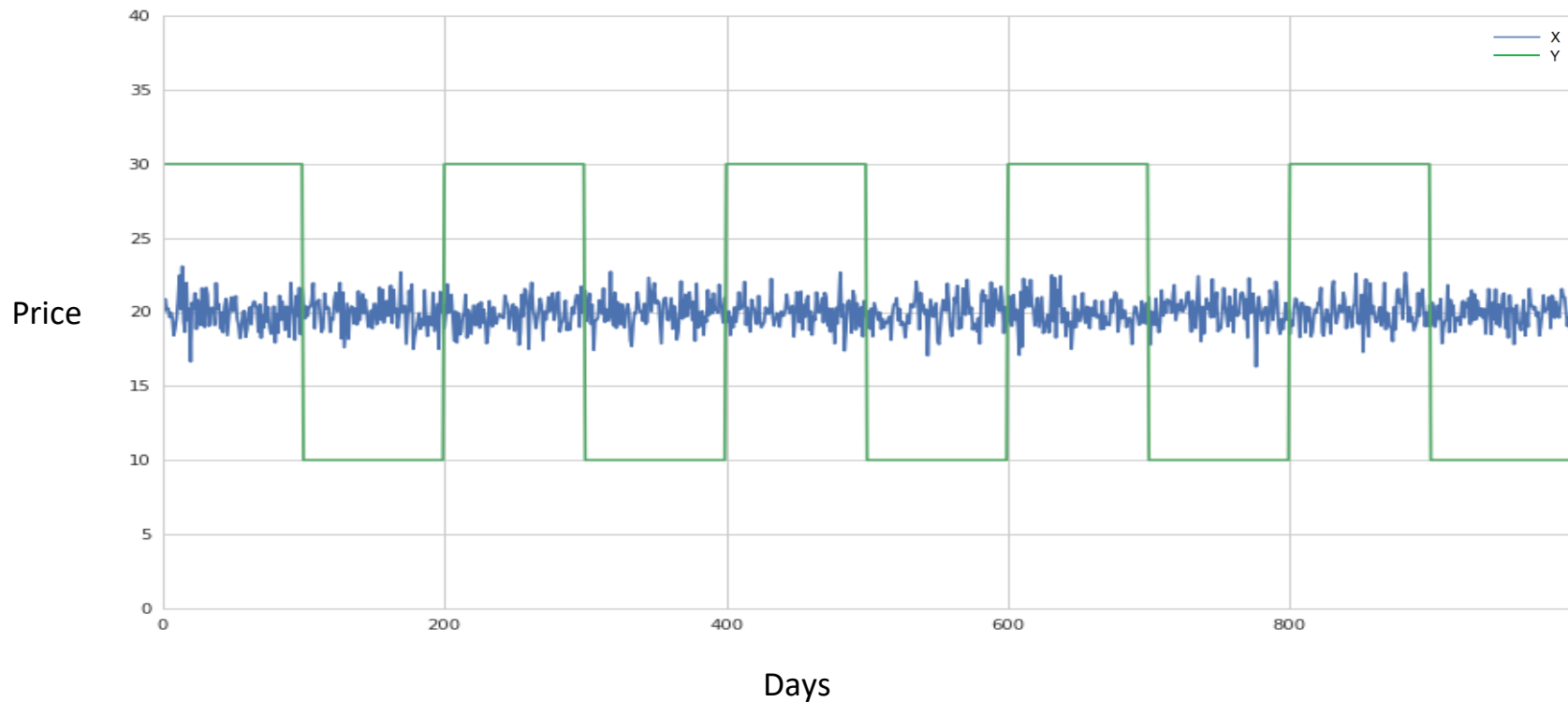
Correlation

- It is whether the two instruments returns move in same direction
- Short-term or long-term

Correlation without Cointegration



Cointegrated without Correlation



Pairs Selection

Why Pair Trading Works?

- Cointegrated pair are assumed to be mean reverting in nature
- Once the spread diverges from its mean, the probability of reversal increases
- With the chosen pair from same industry and sector, extreme divergence from mean is rare

- Pair Selection Criteria
 - Qualitative Selection (ETFs & Currencies)
 - Exposed to common economic factors
 - Qualitative Selection (Stocks)
 - Same Sector
 - Similar Market Capitalization
 - Similar Ratios

Note: Many choices of pairs but easy to lose cointegration

- Cointegration Check
 - ADF Test

- [Example 1:](#) Both (EWA) Australia and (EWC) Canada are dominated by commodities stocks
- [Example 2:](#) ACC and Ambuja

Criteria	ACC	Ambuja
Sector	Cement & Cement Products	Cement & Cement Products
Market Capitalization (In INR Cr.)	32,805	53,235
ROE (3 Years)	9.88%	8.74%

Risk Management

Case Study: LTCM

- Statistical Arbitrage is **not a risk-free strategy**
- Rather than converging, the spread can begin to diverge (drift apart)
 - *The spread picks up trend rather than mean-reverting and the cointegration is broken*
- An event in a security can trigger extreme movement in the spread

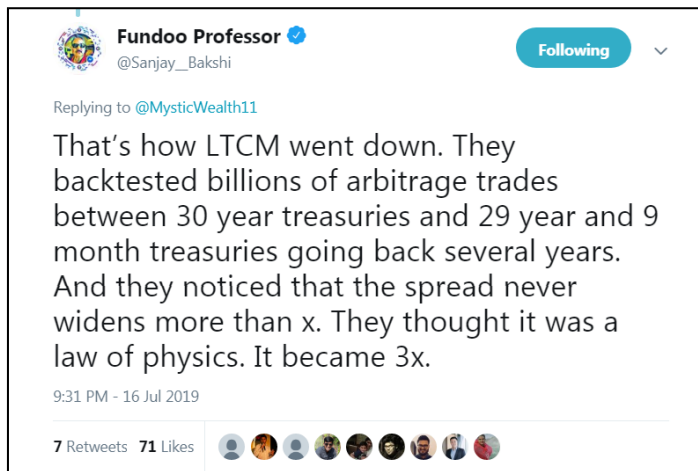
Strict risk management is required to handle adverse situations once the mean-reverting behavior is invalidated

- Define strict stop-loss and profit mechanism before entering trade
- It's a good practice to allocate fund to different pair portfolio rather than one single trade
- Combine with momentum strategies

How to place the stop-loss in these strategies?

Recipe for disaster

- Overreliance on backtest
- High leverage



Summary

- ✓ Stationarity and Cointegration are important for profitable mean reversion strategy
- ✓ ADF Test is used to determine whether two price series is stationary/cointegrated or not
- ✓ Stock pairs are quite unstable w.r.t. cointegration
- ✓ ETFs and currency pair are good candidates for pair trading

- Index arbitrage strategy
- Johansen test
- Common pitfalls in backtesting
- Half-life

- Ernest P. Chan “Algorithmic trading winning strategies and their rationale”
- Gatev, Evan, William N. Goetzmann, and K. Geert Rouwenhorst, “Pairs Trading: Performance of a Relative-Value Arbitrage Rule,” *Review of Financial Studies* (2006): 797-827
- Vidyamurthy, Ganapathy, *Pairs Trading: Quantitative Methods and Analysis* (New Jersey: John Wiley & Sons, Inc., 2004)
- Wooldridge, Jefferey M., *Introductory Econometrics, A Modern Approach, Third Edition* (Ohio: Thomson South-Western, 2006)
- Screener.in

- [Email](#)
- [LinkedIn](#)
- [Quora](#)

- Half-life
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1404905
https://en.wikipedia.org/wiki/Ornstein%E2%80%93Uhlenbeck_process
- The Cointegration Alpha
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=315619
- Statistical Arbitrage in the U.S. Equities Market
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1153505
- Optimal Mean Reversion Trading with Transaction Costs and Stop-Loss Exit
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2222196
- Implementation of Pairs Trading Strategies
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1594066
- Quantitative Spread Trading on Crude Oil and Refined Products Markets
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1932471
- Pairs Trading on International ETFs
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1958546

- Pairs Trading: Performance of a Relative Value Arbitrage Rule
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=227415
- Cointegration Trading with Log Prices vs. Prices
https://www.jstor.org/stable/4480875?seq=1#page_scan_tab_contents
<http://epchan.blogspot.com/2013/11/cointegration-trading-with-log-prices.html>
- Johansen Test in Python
<https://www.quantinsti.com/blog/johansen-test-cointegration-building-stationary-portfolio/>
- A Simplified Approach to Understanding the Kalman Filter Technique
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=715301
- Application of Machine Learning Tools in Predictive Modeling of Pairs Trade in Indian Stock Market
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3159868

Thank you!